

Lupus Nephritis

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DEFINITION

Lupus nephritis (LN), the prototypical immune complex glomerulonephritis (GN), occurs commonly in patients with systemic lupus erythematosus (SLE) and can be very severe. SLE is defined by specific clinical and laboratory features (Table 27.1).¹ To be considered as having SLE, a patient must demonstrate antinuclear antibody (ANA) positivity of 1:80 or more using a Hep2 cell or equivalent assay. Other clinical and serologic parameters are assigned points and, in addition to a positive ANA, a patient must have sufficient clinical and laboratory findings to achieve a score of at least 10. Importantly, proteinuria and kidney biopsy findings are heavily weighted. A kidney biopsy showing an immune complex GN consistent with an International Society of Nephrology/Renal Pathology Society (ISN/RPS) class III or IV LN will receive a score of 10, so this plus a positive ANA are together sufficient to establish a diagnosis of SLE and LN.

EPIDEMIOLOGY

The incidence and prevalence of lupus and LN are influenced by age, sex, ethnicity, geographic region, diagnostic criteria used, and method of ascertainment. Clinically important kidney disease occurs in about 50% of patients with SLE, mostly between the ages of 15 and 45 years, with women outnumbering men 8:1 to 15:1.² Female predominance is less pronounced in children and older individuals. Among patients with lupus, LN affects both sexes equally, is more severe in children and men, and is less so in older adults. The incidence of LN is about 30% in White, 60% in Black and Hispanic, and 40% to 80% in Asian patients with SLE.^{3,4} The higher frequency of LN in Black patients is independent of socioeconomic status.⁵ Black and Hispanic patients with SLE develop LN earlier and have worse outcomes than White patients. The prevalence of LN in a Medicaid population was 31 in 100,000.⁵

A pragmatic review found the 5- and 10-year cumulative incidence of end-stage kidney disease (ESKD) to be 11% and 19%, respectively,² but this depends on histologic class and the population sampled. The risk for ESKD over 15 years was up to 44% in class IV LN.⁶ In a Medicaid population with a new diagnosis of LN between 2000 and 2010, the 5-year cumulative incidence of ESKD was about 22%.⁷

SLE patients with LN die earlier than those without nephritis and have a standardized mortality ratio of 6 to 6.8 versus 2.4 in lupus without kidney involvement.^{8,9} This increases to 14 for those with chronic kidney disease and 63 for those with ESKD.¹⁰ However, if LN remission is achieved through treatment, 10-year survival doubles to 95%.¹¹

ETIOLOGY AND PATHOGENESIS

Genetics and Environment

A genetic predisposition to SLE is supported by disease clustering in families, twin concordance, racial differences in susceptibility, and the high

frequency of autoantibodies and other autoimmune disorders in healthy family members.¹² In Taiwan the relative risk for lupus in families of those with SLE was 316 for twins, 24 for siblings, 11 for parents, 14 for children of patients, and 4.4 for nonrelated spouses, suggesting that heritability contributed 44% to the risk for SLE, shared environmental exposures contributed 26%, and nonshared environmental exposures 30%.¹³

Homozygous deficiency of early complement cascade components (C1q, C2, C4) is associated with a high risk of SLE, as are certain FcγRIII receptor polymorphisms. Genome-wide association studies have identified over 100 loci associated with an increased risk for SLE. Genetic risk scores are lowest in Europeans, higher in Asians, and highest in those of African ancestry.¹⁴ Genes involved in lupus risk include those affecting B-cell signaling, neutrophil function, interferon regulation, immune-complex clearance, toll-like receptors (TLRs), and the B-cell survival factor BAFF.^{12,15} The BAFF variant results in higher levels of BAFF and contributes to the rationale of treating lupus and LN with a monoclonal antibody against BAFF. A disproportionate number of transcription factors have been identified, suggesting an important contribution of gene dysregulation to lupus.¹⁶ HLA genes are also strongly associated with lupus risk, at least in Europeans.¹⁷ In Whites and Asians, four HLA class II DR allele families confer increased susceptibility or resistance to the development of LN.¹⁸ Independent of causation, other polymorphisms in genes such as *MYH-9*, *ACE*, *TNIP-1*, and *APOL-1* (in Blacks) confer a worse prognosis for LN. Having even one *APOL-1* kidney risk allele was associated with progressive chronic kidney disease (CKD) and more chronicity, especially interstitial fibrosis and tubular atrophy, in nonwhite Brazilian patients.¹⁹

External environmental factors, such as ultraviolet light exposure and smoking, play a role in the onset and exacerbation of SLE and LN.²⁰ Viral infections may be triggers for lupus, but conclusive evidence for a viral pathogenesis of SLE or LN is lacking. Exposure to certain medications (e.g., procainamide, hydralazine, quinidine, and anti-tumor necrosis factor [TNF] biologics) has been linked to SLE or SLE-like syndromes, but LN occurs infrequently in such individuals.

The internal environment also appears to be relevant in SLE given the strong female predominance, disease exacerbations during or shortly after pregnancy, and the effects of hormone treatment and ablation in animal models of LN. The gut microbiome may also contribute to lupus disease pathogenesis and activity. For example, orthologs of the RNA binding autoantigen Ro60 have been found in gut, skin, and oral commensal bacteria.²¹ Autoantibodies to Ro60 are seen early in patients who develop SLE. The Ro60 orthologs could trigger autoimmunity in genetically susceptible individuals. Dysbiosis of the gut microbiome occurs in lupus patients. In some patients with active LN, circulating antibodies against overrepresented bacterial species have been found.²² Anti-double stranded DNA (dsDNA) antibodies can recognize an antigen from these bacteria, again supporting a link

antinuclear cytoplasmic antibodies (ANCA), which occur more often in proliferative than membranous LN and are associated with worse baseline kidney function.²⁶

PATHOGENESIS OF KIDNEY INJURY IN LUPUS NEPHRITIS

The hallmark of LN is the accumulation of immune complexes in glomeruli.²⁷ Patients with LN have autoantibodies against dsDNA, Sm antigen, C1q, nucleosomes, and other antigens. There is direct binding of dsDNA antibodies to the glomerular basement membrane (GBM), and cross-linking of positively charged nucleosome components such as chromatin between autoantibodies and GBM.²⁷ In proliferative LN, immune complexes are found in the subendothelial space, whereas in membranous LN, they localize to the subepithelial space.

Glomerular immune complexes, especially in the subendothelial space, activate proinflammatory mechanisms, including the complement pathway, leukocyte Fc receptors, cytokines that regulate cell proliferation and matrix formation, and procoagulant factors.^{27–29} Nucleosomes also can activate resident dendritic cells through binding to TLR-2 and -9.³⁰ These activated pathways result in kidney damage, intraglomerular hypertension and coagulation, and leukocyte infiltration with release of proteolytic enzymes. Subepithelial immune complexes are associated with less inflammation but increased production of GBM components and more podocyte injury.

T cells contribute to the progression of LN by facilitating B cell differentiation and expansion. Additionally, T helper cell (Th1) cytokines are overexpressed in kidneys of patients with LN and promote intrarenal inflammation by activation of macrophages, complement, and the Fc receptor pathway. Kidney biopsies of LN patients also contain Th17 cells, which secrete interleukin-17 (IL-17), which is thought to sustain kidney inflammation by driving T cells away from maturing into a regulatory phenotype (CD4⁺CD25^{hi}FoxP3⁺) capable of suppressing autoantibody production and attenuating the immune response.³¹ CD8⁺ T cells may gradually lose effector function and express inhibitory receptors under persistent antigen exposure and become “exhausted.” Exhaustion may, however, explain, why LN patients whose peripheral T cells displayed an exhausted phenotype had a nonrelapsing disease course.³²

In SLE and LN, neutrophils undergo a form of cell death called NETosis, in which a chromatin meshwork (or NET) is released.³⁰ These NETs are a source of autoantigens and are not properly degraded in lupus patients. In addition, lupus patients have an increased number of low-density granulocytes that are more susceptible to NETosis. NET material has been found in LN biopsy samples, and the subset of SLE patients with LN often do not degrade NETs well. The NETs induce production of IFN- α by plasmacytoid dendritic cells, which are also found in LN kidneys.³⁰

Within the tubulointerstitial compartment, T and B cells are often found in close proximity to each other and appear to be interacting, in some cases even forming germinal centers.³³ Interstitial B cells aggregating with T cells may show clonal expansion and somatic hypermutation, suggesting intrarenal autoantibody production against kidney-specific antigens, such as vimentin. Such interactions may contribute to interstitial inflammation, a major determinant of long-term kidney survival.³³

Single-cell RNA sequencing (scRNA-seq) identified 21 clusters of immune cells in the LN kidney,³⁴ all displaying an interferon signature, much as is seen in the peripheral blood of patients with SLE and LN. Of the three clusters of CD8⁺ T cells found in the kidney, none had an exhaustion signature, in contrast to that found in the peripheral blood of some patients. This demonstrates the complex and sometimes unexpected relationships among intrarenal immune cells and between tissue and peripheral blood immune cells.

TABLE 27.2 Frequency of Kidney Manifestations in Patients With Lupus and Kidney Involvement

Manifestation	Prevalence (%)
Proteinuria	100
Nephrotic syndrome	45–65
Hematuria	
Microscopic	80
Gross	1–2
Red blood cell casts	10
Cellular casts	30
Reduced kidney function	40–80
Rapidly progressive glomerulonephritis	10–20
Acute kidney injury	1–2
Hypertension	15–50
Hyperkalemia	15

CLINICAL MANIFESTATIONS

Immune complex kidney involvement (LN) is usually heralded by proteinuria and active urinary sediment with microhematuria, acanthocytes, and erythrocyte casts (Table 27.2). More severe LN, usually with proliferative histology, may present with hypertension and a decline in the glomerular filtration rate (GFR). Less frequently, kidney disease in lupus presents as a tubulointerstitial disorder such as renal tubular acidosis (see Chapter 13), isolated interstitial nephritis, or as a thrombotic microangiopathy with or without an antiphospholipid antibody syndrome (see Chapters 29 and 30).

Extrarenal Manifestations

Patients with active SLE often present with nonspecific complaints of malaise, low-grade fever, poor appetite, and weight loss. Other common features include patchy alopecia, oral or nasal ulcerations, arthralgias, nondeforming arthritis, and a variety of skin findings, including photosensitivity, Raynaud’s phenomenon, and a “butterfly” (malar) facial rash. Livedo reticularis is seen in up to 15% of cases and may be associated with miscarriage, thrombocytopenia, and antiphospholipid antibodies. Neuropsychiatric involvement presents with headache, nerve palsies, psychoses, or frank coma. Serositis, in the form of pleurisy or pericarditis, affects up to 40% of patients. Pulmonary hypertension can develop silently as a result of multiple pulmonary emboli or intravascular coagulation in association with antiphospholipid antibodies or may be caused by nonthrombotic pulmonary arterial disease. Mitral valve prolapse, and less commonly Libman-Sacks endocarditis, may both be seen in SLE. Splenomegaly and lymphadenopathy are present in about one-fourth of patients, and hematologic abnormalities can affect all three cell lines. SLE-associated anemia can result from impaired erythropoiesis, autoimmune hemolysis, or bleeding. Thrombocytopenia and leukopenia may occur as part of lupus or as complications of therapy. Thrombotic events are common and should prompt a search for antiphospholipid antibodies and other procoagulant abnormalities.

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

Although the diagnosis of lupus may be obvious in a young woman with classic manifestations and serologic markers, less typical presentations are common and often result in diagnostic delay. This is, in

part, a result of the varied features of the disease and because signs and symptoms evolve over time.

Several autoimmune diseases mimic the extrarenal manifestations of SLE, including fibromyalgia, Sjögren syndrome, thrombotic microangiopathies, primary antiphospholipid syndrome, dermatomyositis, systemic sclerosis, and mixed connective tissue disease. Adding complexity to diagnosis, many of these autoimmune diseases can involve the kidney and, in some cases, are associated with GN. Conversely, several common forms of GN must be distinguished from LN because of similar clinical features, including vasculitides (see [Chapters 24 and 26](#)), bacterial endocarditis, and cryoglobulinemia. ANCAs of uncertain significance are detected in 20% of LN patients.

Immunologic Tests in Lupus

ANAs are found in more than 90% of untreated patients with lupus. Although highly sensitive, neither the presence nor pattern (e.g., diffuse, speckled) are specific for SLE. Autoantibodies against dsDNA are more specific but less sensitive, being present in 75% of untreated lupus patients. Whereas high titers of anti-dsDNA antibodies correlate with the presence of SLE and are often used to follow the course of LN, antibodies to single-stranded DNA (ssDNA) are found in many rheumatologic conditions and do not correlate with the course of LN. Sm antibodies are strongly associated with the diagnosis of lupus and LN but are present in only about 25% to 30% of patients. Antibodies to C1q (anti-C1q) have been more closely associated with the activity of LN than anti-dsDNA antibodies and may have a prognostic role in the follow-up.³⁵

Serum levels of total hemolytic complement and complement components C3 and C4 are often depressed in untreated SLE and especially in LN. In general, both C3 and C4 are depressed. Preferential C4 depression in lupus patients may reflect activation of the classic complement pathway. In some cases, low C4 with normal C3 in a patient with lupus may reflect genetic C4 deficiency or the presence

of cryoglobulins. Preferential depression of C3 is observed in patients with postinfectious glomerulonephritis (PIGN) and the C3 glomerulopathies (see [Chapters 23 and 57](#)) rather than in SLE/LN.

KIDNEY BIOPSY IN SYSTEMIC LUPUS ERYTHEMATOSUS

Although lupus nephritis may be suspected based on clinical symptoms and laboratory markers, a kidney biopsy is required for confirmation, subclassification, prognosis, and management decisions. A biopsy should be done in patients with a persistent urine protein/creatinine ratio greater than 0.5 g/g (equivalent to proteinuria of about 0.5 g/day), with or without an active urine sediment (i.e., glomerular hematuria and/or leukocyturia in the absence of urinary tract infection). Although a fall in GFR because of LN is usually accompanied by proteinuria and hematuria, a biopsy should be done if kidney function declines in an SLE patient in the absence of urinary abnormalities if no other reason (e.g., infection, nephrotoxic medications) for the falling GFR can be found.

The proteinuria cutoff of 0.5 g/g is not an absolute threshold. A kidney biopsy should be considered for individual patients in the context of their overall disease characteristics. For example, proliferative LN can be found in lupus patients with proteinuria less than 0.5 g/g who have glomerular hematuria with or without impaired kidney function. In one cohort, over 80% of such patients had class III or IV LN on kidney biopsy.³⁶

After LN treatment, a second kidney biopsy can be considered for patients who have not responded adequately, have persistently elevated markers of kidney disease activity, or who deteriorate and in whom future treatment is unclear.

PATHOLOGY

Glomerular involvement is the dominant finding in LN. LN is currently classified by the revised ISN/RPS system ([Table 27.3](#))³⁷ and is based mainly

TABLE 27.3 ISN/RPS Classification of Lupus Nephritis and Associated Clinical Correlations

Class	Definition	Urine Findings	Clinical Findings
I: Minimal mesangial LN	Normal glomeruli by LM but mesangial immune deposits by IF	Usually unremarkable	None relevant to kidney; excellent kidney prognosis ^a
II: Mesangial proliferative LN	Mesangial hypercellularity (4 or more nuclei fully surrounded by matrix in the mesangial area excluding the hilar region) with mesangial immune deposits	Microscopic hematuria; proteinuria, if present, is usually low grade	Preserved kidney function; hypertension infrequent; excellent kidney prognosis ^a
III: Focal LN	Segmental or global, active endocapillary or extracapillary, or inactive glomerulonephritis affecting <50% of glomeruli with mesangial and subendothelial immune deposits	Microscopic hematuria; proteinuria	Hypertension possible; kidney failure and nephrotic syndrome not unusual; variable kidney prognosis
IV: Diffuse LN	Segmental or global active endocapillary or extracapillary, or inactive glomerulonephritis affecting ≥50% of glomeruli with mesangial and subendothelial immune deposits	Microscopic hematuria; proteinuria	Hypertension; kidney failure and nephrotic syndrome frequent; variable kidney prognosis
V: Membranous LN	Glomerular basement membrane thickening with subepithelial and mesangial immune deposits; can occur alone or in combination with class III or IV	High-grade proteinuria; microscopic hematuria possible	Preserved kidney function; nephrotic syndrome common; kidney prognosis good ^a Anti-PLA2R antibody negative
VI: Advanced sclerosing LN	≥90% of glomeruli globally sclerosed without residual disease activity	Microscopic hematuria; proteinuria not unusual	Kidney failure expected

^aAs long as there is no transformation to a proliferative class.

IF, Immunofluorescence microscopy; ISN/RPS, International Society of Nephrology/Renal Pathology Society; LM, light microscopy; LN, lupus nephritis.

on glomerular histology (examples are shown in Figs. 27.1 through 27.5). Less common glomerular pathologies omitted from the ISN/RPS system include lupus podocytopathy (see later discussion) and pauci-immune crescentic GN (analogous to ANCA-associated renal vasculitis).

Although IgG is generally the dominant glomerular immunoglobulin in LN, IgA, and IgM, along with the complement components C1q and C3, are often co-deposited, giving the “full-house” pattern that is highly suggestive of LN. Strong glomerular C1q staining is also

suggestive of LN. Fibrin staining, corresponding to active lesions, is often noted in the glomerular tuft, especially in crescents.

On electron microscopy (EM), the distribution of electron dense, immune complex deposits corresponds to the pattern of immunoglobulins seen by immunofluorescence or immunohistology. Some electron-dense deposits have an organized substructure known as *fingerprinting*, corresponding to the presence of curvilinear microtubular or fibrillar structures. Tubuloreticular inclusions, which are tubular structures located in the endoplasmic reticulum of kidney endothelial cells, are often found in biopsy specimens of LN patients and are thought to reflect a high interferon milieu.

Tubulointerstitial and Vascular Disease

The original ISN/RPS classification does not assess the kidney tubulo-Interstitial or vascular compartments, even though interstitial and vascular damage contribute to poor kidney outcomes.³⁸ The updated ISN/RPS classification includes activity and chronicity scores for glomerular and tubulointerstitial lesions, and vascular lesion grading along the lines of the Banff classification for transplant biopsies.³⁷

About 50% of patients with LN, predominantly those with proliferative glomerular lesions, have immune aggregates along tubular basement membranes. Interstitial inflammatory cell infiltrates, including T cells, B cells, and monocytes, are frequently found, and tubulitis can be seen in active disease (Fig. 27.6). In chronic disease, the interstitium is often expanded by fibrosis and sparser infiltrates. Infrequently, tubulointerstitial nephritis is seen in the absence of glomerular disease and may result in acute kidney injury (AKI) or renal tubular acidosis.

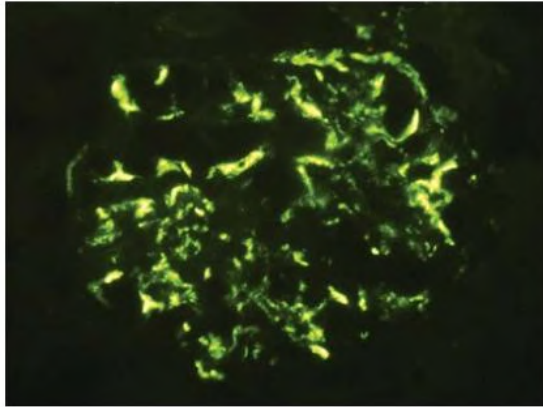


Fig. 27.1 International Society of Nephrology/Renal Pathology Society Class I: Minimal Mesangial Lupus Nephritis. Light microscopy is normal (*not shown*), but direct immunofluorescence shows C1q deposits in the mesangial area. No capillary walls are involved. (Magnification $\times 400$.)

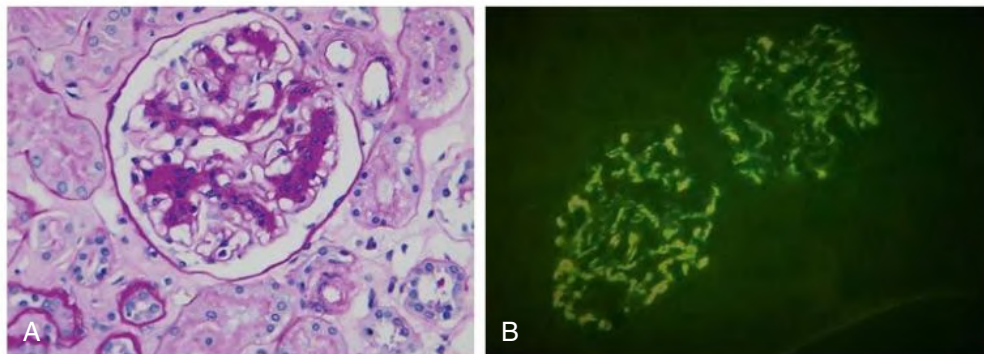


Fig. 27.2 International Society of Nephrology/Renal Pathology Society Class II: Mesangial Proliferative Lupus Nephritis. (A) Mesangial expansion with hypercellularity, normal capillary walls, and no endocapillary hypercellularity (PAS stain, $\times 400$). (B) Diffuse mesangial immunoglobulin G deposits shown by direct immunofluorescence ($\times 200$).

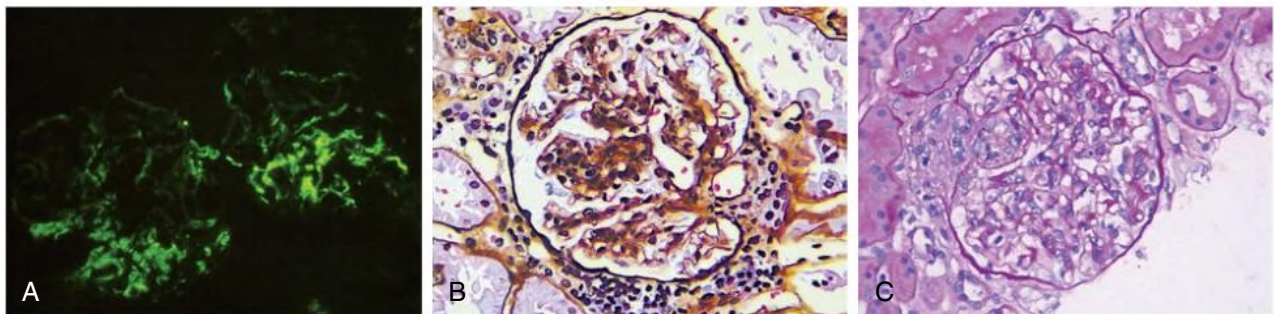


Fig. 27.3 International Society of Nephrology/Renal Pathology Society Class III: Focal Proliferative Lupus Nephritis. (A) Segmental deposits of immune complexes in the mesangial areas and in some capillary walls (C1q, direct immunofluorescence, $\times 200$). (B) Glomerulus showing segmental endocapillary hypercellularity (silver methenamine stain, $\times 400$). (C) Area of endocapillary hypercellularity with karyorrhexis and a segmental fibrocellular crescent with focal disruption of Bowman's capsule (PAS stain, $\times 400$).

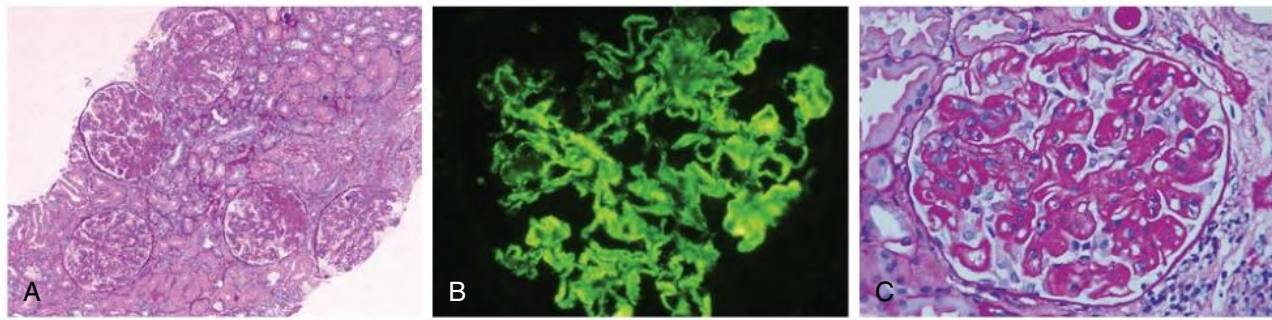


Fig. 27.4 International Society of Nephrology/Renal Pathology Society Class IV: Diffuse Proliferative Lupus Nephritis. (A) Active diffuse proliferative lupus nephritis showing the presence of endocapillary hypercellularity with lobular accentuation of the glomerular capillaries and hyaline deposits that compromise the majority of the capillary walls (PAS stain, $\times 40$). (B) Prominent global subendothelial deposits, accompanied by immune deposits in the mesangium (direct immunofluorescence for immunoglobulin, $\times 400$). (C) Extensive subendothelial immune deposits thickening almost all the capillary walls, generating a wire loop appearance and hyaline thrombi that occlude capillary lumens (PAS stain, $\times 400$).

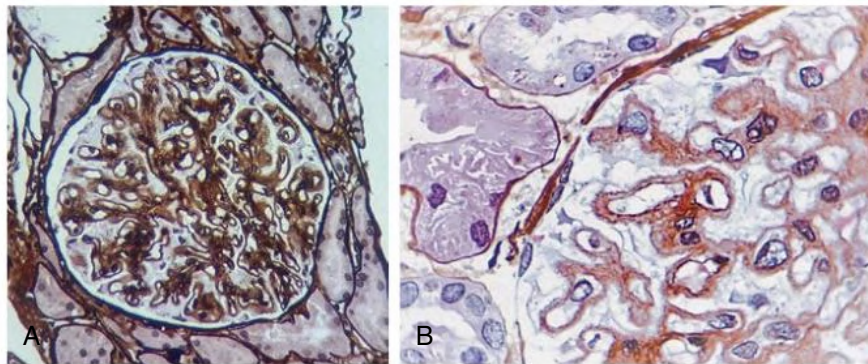


Fig. 27.5 International Society of Nephrology/Renal Pathology Society Class V: Membranous Lupus Nephritis. (A) Global subepithelial deposits with spike formation (silver methenamine stain, $\times 400$). (B) Presence of extensive subepithelial deposits and segmental subendothelial deposits that generate double contours and indicates membranous lupus nephritis with a proliferative component (class V + III/IV) (silver methenamine stain, $\times 400$).

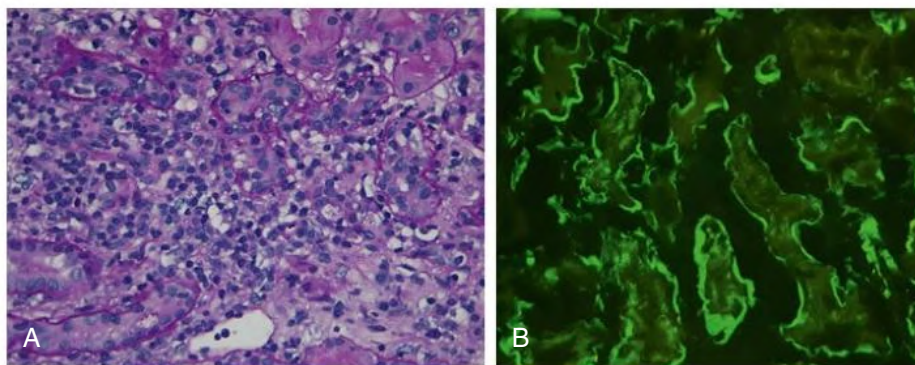


Fig. 27.6 Tubular Injury in Lupus Nephritis. (A) Interstitial lupus nephritis: mononuclear cells invading and destroying the tubular basement membranes (tubulitis and tubulorrhexis). Interstitial nephritis should not be related to chronic lesions such as tubular atrophy or glomerular scarring to contribute to the activity index. (PAS stain, $\times 400$). (B) Direct immunofluorescence showing immunoglobulin G deposits in the tubular basement membrane ($\times 400$). Such tubular basement membrane aggregates are common in lupus nephritis, being found in 60% to 65% of biopsy specimens overall and with increasing frequency from class II (20%) to class IV (75%).

Vascular lesions are common in LN (Fig. 27.7). Five main pathologic types of kidney microvascular injury are seen: (1) vascular immune-complex deposits, (2) arteriosclerosis, (3) thrombotic microangiopathy, (4) noninflammatory necrotizing vasculopathy and, less frequently, (5) true vasculitis.³⁹

Transformation of Histologic Appearance and “Silent” Lupus Nephritis

The ISN/RPS class found for LN is not fixed for an individual’s entire clinical course. Those treated for proliferative (class III/IV) LN may transform to a less serious histology, such as class II, or have resolution

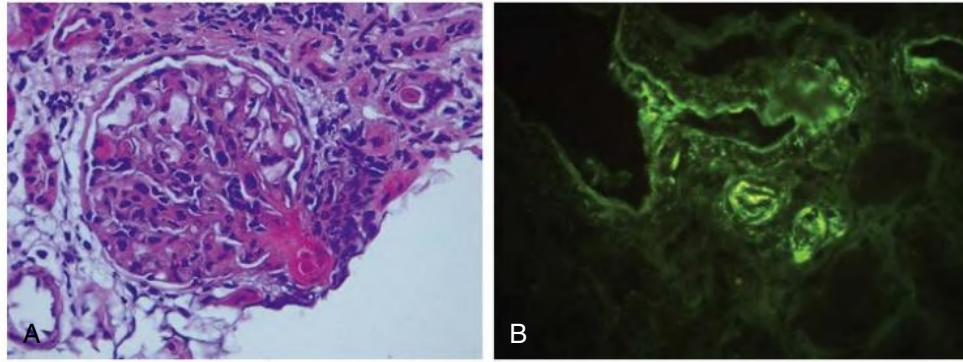


Fig. 27.7 Vascular Injury in Lupus Nephritis. (A) Lupus vasculopathy: luminal narrowing of arterioles by intramural immune deposits with fibrinoid changes but without inflammation of the vessel wall. (B) Demonstration of C1q deposits along the arterial walls, confirming that the hyaline deposits seen by light microscopy correspond to immune complexes and the diagnosis of lupus vasculopathy (direct immunofluorescence, $\times 400$).

TABLE 27.4 Modified NIH Lupus Nephritis Activity and Chronicity Scoring System^a

Semiquantitative Lesion Score ^a	ACTIVITY INDEX						CHRONICITY INDEX			
	Cellular/Fibrocellular Crescents	Fibrinoid Necrosis	Neutrophils/Karyorrhexis	Endocapillary Hypercellularity	Hyaline Deposits	Interstitial Inflammation	Total Glomerulosclerosis Score (Segmental or Global)	Fibrous Crescent	Tubular Atrophy	Interstitial Fibrosis
None	0	0	0	0	0	0	0	0	0	0
Mild (<25%)	2	2	1	1	1	1	1	1	1	1
Moderate (25%–50%)	4	4	2	2	2	2	2	2	2	2
Severe (>50%)	6	6	3	3	3	3	3	3	3	3

^aMaximum activity index is 24; maximum chronicity index is 12.

NIH, National Institutes of Health.

of inflammatory lesions with scarring and move from an active to a chronic histologic pattern. Conversely, patients initially diagnosed with class II or V LN may transform to proliferative LN, often heralded by increasing proteinuria and activity of the urine sediment.⁴⁰

Kidney biopsies have been done in patients with SLE but no clinically obvious kidney involvement⁴¹; glomerular immune complexes were seen in some cases, most often consistent with class I/II LN but sometimes consistent with class III, IV, or V LN. This has been termed *silent LN* and may represent a preclinical stage in the evolution of LN.

Clinical, Laboratory, and Histopathologic Correlations and Outcomes

Kidney biopsy remains the gold standard for diagnosis and subsequent management of LN. Table 27.3 summarizes the typical clinicopathologic correlations of LN. Lupus membranous nephropathy is distinguished from the most frequent type of primary membranous nephropathy by the absence of circulating anti-PLA2r autoantibodies and by the presence of all IgG isotypes in the intrarenal immune deposits, although restriction to the IgG4 isotype has been rarely observed in LN patients.

The importance of the kidney biopsy is underscored by patients with SLE and kidney injury who do not have classic LN. For example, up to 24% of patients may have a renal thrombotic microangiopathy, either alone or associated with immune-complex GN.⁴² A

small (1.3%) but important subset of patients with SLE present with nephrotic syndrome, with or without AKI, but their kidney biopsies either mimic minimal change disease (MCD) or show mild mesangial expansion, proliferation and segmental sclerosis, mimicking focal segmental glomerulosclerosis (FSGS).⁴³ EM shows diffuse podocyte effacement, without immune complexes, or with mesangial deposits but no peripheral capillary deposits of immunoglobulins or complement components. These patients, although not classified by ISN/RPS, have lupus podocytopathy and like MCD or FSGS, most respond to corticosteroids alone.⁴⁴ However, in a Chinese study, the risk for relapse exceeded 50%, and at repeat kidney biopsy, half of the relapsed patients had transformed to class IV or V LN.⁴⁴ Although these results may not be generalizable to a non-Asian population, they underscore the importance of histology-driven treatment decision analysis.

Long-Term Prognosis and Kidney Histology

Features of reversible (active) or irreversible (chronic) damage on kidney biopsy are captured by the National Institutes of Health (NIH) activity and chronicity indices now adapted to the proposed revisions of the ISN/RPS classification (Table 27.4; Fig. 27.8).³⁷ Similar to other types of GN, long-term kidney prognosis is poor if the biopsy sample shows extensive glomerulosclerosis or interstitial fibrosis and tubular atrophy. In patients with less extensive scarring, the activity and chronicity indices at diagnostic biopsy may not correlate with the course of

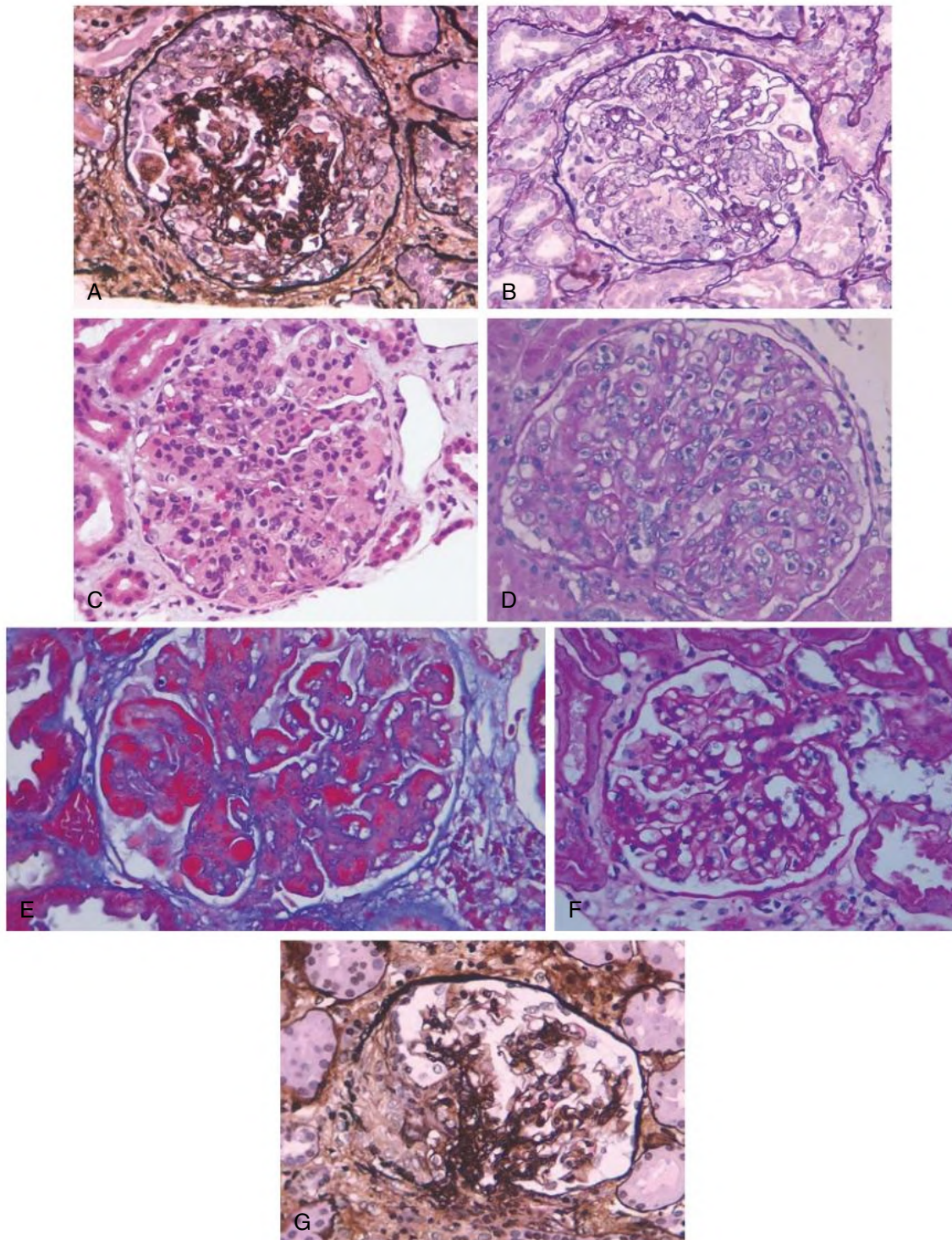


Fig. 27.8 Histologic Components of the Activity and Chronicity Indices. (A) Fibrocellular crescent. A crescent is defined by the presence of extracapillary hypercellularity, composed of more than two cell layers that make up 10% or more of the circumference of Bowman's capsule. Fibrocellular crescents have 25% to 75% cells and fibrin, and the remainder of fibrous matrix, and should be considered in the activity index (silver methenamine stain, $\times 400$). (B) Fibrinoid necrosis. Fibrin associated with glomerular basement membrane disruption; this lesion does not require the presence of karyorrhexis (silver methenamine stain, $\times 400$). (C) Neutrophils/karyorrhexis. Presence of neutrophils in the capillary lumina and karyorrhexis; defined as the presence of apoptotic, pyknotic, and fragmented nuclei (H&E, $\times 400$). (D) Endocapillary hypercellularity. Increased number of infiltrating monocytes causing narrowing of the glomerular capillary lumina (PAS stain, $\times 400$). (E) Hyaline deposits. Wire loop lesions and hyaline thrombi that represent the presence of massive subendothelial deposits that occlude the capillary lumina (Masson's trichrome stain, $\times 400$). (F) Glomerulosclerosis. Segmental sclerosis with fibrous adhesions to Bowman's capsule (PAS stain, $\times 400$). For the chronicity index, segmental and global glomerulosclerosis are considered together. (G) Fibrous crescent. Extracapillary proliferation with more than 75% fibrous matrix and less than 25% cells and fibrin (Silver methenamine stain, $\times 400$).

LN.⁴⁰ In contrast, persistent activity (activity index > 2) despite treatment on repeat kidney biopsy is associated with a worse long-term kidney prognosis, as is high chronicity (chronicity index > 6).⁴⁰

TREATMENT

The management of patients with SLE (including those with LN but no systemic disease) should include an antimalarial agent such as hydroxychloroquine. Antimalarials are TLR antagonists (specifically TLR-7, -8, and -9) and reduce flare rates (including LN flares), decrease organ damage, and enhance responsiveness to immunosuppression, among other beneficial effects.⁴⁴ Patients should be monitored for retinal toxicity.

The treatment of patients with active proliferative LN is traditionally divided into induction and maintenance phases, although these distinctions are blurred by the design of novel therapeutic trials (discussed later). The *initial* or *induction phase* may be thought of as the first step in controlling kidney inflammation to allow healing of kidney injury. This translates clinically into stabilization or improvement of kidney function, attenuation of proteinuria, and reduction in urine sediment activity. The *maintenance phase* focuses on consolidating clinical remissions and preventing relapses while reducing exposure to potentially toxic medications.

The ISN/RPS biopsy classification should guide initial therapy (see Table 27.3). In general, the immunosuppressive treatment of extrarenal lupus manifestations is sufficient for class I and II LN. The combination of high-dose glucocorticoids plus an immunosuppressive agent is mainly used for patients with active focal and diffuse proliferative LN (classes III and IV) and membranous lupus (class V).

A complete kidney response is achieved in only 30% to 40% of LN patients by 12 months.⁴⁵ Most studies define complete kidney response as a reduction in proteinuria to less than 0.5 g/day or a urine protein to creatinine ratio less than 0.5 g/g, absence of glomerular hematuria and red blood cell casts, and normal or stable GFR. However, long-term kidney survival is best predicted by reducing proteinuria to less than 0.7 to 0.8 g/day after 1 year of treatment.^{46–48} Importantly this cutoff is starting to be incorporated into phase III clinical trials of novel LN therapies.⁴⁹ The term *partial response* requires a 50% reduction in proteinuria to subnephrotic levels, and stability or improvement in GFR and is usually achieved before complete response criteria are met.

In addition to immunosuppressive regimens, the kidney protective measures outlined in Chapter 82 should be used as appropriate.

Proliferative Lupus Nephritis: Initial Treatment Glucocorticoids

The administration of glucocorticoids is fundamental to the treatment of LN. The goal of initial steroid therapy is to quickly control intrarenal inflammation and halt ongoing injury, thereby allowing time for coadministered immunosuppressive agents to attenuate the autoimmune mechanisms that initiated and perpetuate LN. Most regimens start with one to three consecutive intravenous (IV) pulses of methylprednisolone at up to 1000 mg/pulse, followed by oral prednisone or prednisolone, often starting at 1 mg/kg/day with slow tapering. There is considerable variability in cumulative glucocorticoid dosing.

There has been a concerted effort to reduce the cumulative steroid burden for LN patients. The original Euro-Lupus Nephritis Trial (ELNT) initiated oral prednisolone at 0.5 mg/kg/day for 1 month, followed by tapering after three 750 mg pulses of methylprednisolone.⁵⁰ The RITUXILUP trial used rituximab plus two 500 mg pulses of IV methylprednisolone followed by maintenance with mycophenolate mofetil (MMF) in the absence of oral steroids.⁵¹ After a median of 36 weeks, 72% of patients had achieved a complete response. During

follow-up, 22% of patients had a LN flare 65 weeks (median) after remission and 64% were managed without oral steroids. A phase II study compared the novel calcineurin inhibitor (CNI) voclosporin to placebo on a background of immunosuppression with MMF and low-dose glucocorticoids.⁵² Patients were given two IV pulses of 500 mg methylprednisolone, followed by 20 to 25 mg/day prednisone that was tapered to 2.5 mg/day by 4 months. A complete kidney response was achieved by 49% of patients in the voclosporin group and 24% of patients in the placebo group after 48 weeks. Taken together, high-dose steroids for all LN patients may not be necessary, oral steroid dosing may be lowered after IV pulses of methylprednisolone, and some therapeutic regimens could be steroid-sparing.

Current LN guidelines recommend initiating steroid treatment with IV methylprednisolone at a total cumulative dose of 250 to 2500 mg (depending on disease severity), followed by oral prednisone at 0.3 to 1 mg/kg/day, and tapered to 7.5 mg/day or less after 3 to 6 months.^{53,54}

Immunosuppressive Agents

Long-term kidney function is better preserved, and LN relapses after 3 to 5 years are reduced if glucocorticoids are combined with cyclophosphamide during initial therapy.⁵⁵ Both daily oral and IV pulses of cyclophosphamide are effective in LN, although IV cyclophosphamide given as 6-monthly pulses of 0.5 to 1 g/m² (NIH protocol) has been the standard of care for several years. The ELNT trial compared low-dose IV cyclophosphamide (500 mg/infusion) bimonthly for 3 months to the NIH protocol. Similar efficacy with less toxicity and fewer infections, and similar 5- and 10-year kidney outcomes were demonstrated.^{50,56} The Euro-Lupus low-dose IV cyclophosphamide regimen achieved similar rates of remission induction in Black, Hispanic, and Southeast Asian patients.^{57,58} Therefore, LN guidelines now recommend low-dose cyclophosphamide as the first choice if a cyclophosphamide-based regimen is to be used.⁵³ However, because low-dose cyclophosphamide studies have not included patients with serum creatinine levels 2.5 mg/dL or higher or those with extensive crescents or necrosis on kidney biopsy, high-dose IV cyclophosphamide remains an option for such patients.^{57,58}

Oral MMF plus glucocorticoids for 6 months followed by maintenance therapy is as effective as 6-monthly pulses of cyclophosphamide (NIH protocol) plus glucocorticoids followed by maintenance therapy.⁴⁵ MMF also showed similar treatment response rates to the Euro-Lupus cyclophosphamide regimen at 24 weeks.⁵⁸ High-dose (NIH protocol) cyclophosphamide has been favored in patients presenting with marked kidney impairment or severe proliferative LN on biopsy, but there are no convincing differences in outcome between MMF and cyclophosphamide induction for such patients.⁵⁹ No improvement in overall mortality or severe infection has been seen with MMF; gastrointestinal disturbances are more frequent, but the risk for amenorrhea is lower than with cyclophosphamide.

Optimal MMF dosing and adjustment for different patient subgroups remains unclear. Although concentration-time curve (AUC_{0–12}) of the active MMF metabolite mycophenolic acid (MPA) may provide accurate overall measurements,⁶⁰ AUC_{0–12} measurement is expensive and difficult to apply clinically. MPA trough levels correlate poorly with total drug exposure by AUC. A 4-hour AUC (AUC_{0–4}) has been proposed to follow MMF exposure, but targets for efficacy and toxicity still need to be established.⁶⁰ To date, a dosing range of 2 to 3 g/day for MMF has been recommended for initial treatment of proliferative LN with dose reduction for intolerance.

Long-term kidney outcomes of patients treated with MMF versus cyclophosphamide are not different, although one study reported a nonsignificant increase in LN flares in patients induced with MMF.⁶¹ Differences in response between ethnic and geographic subgroups

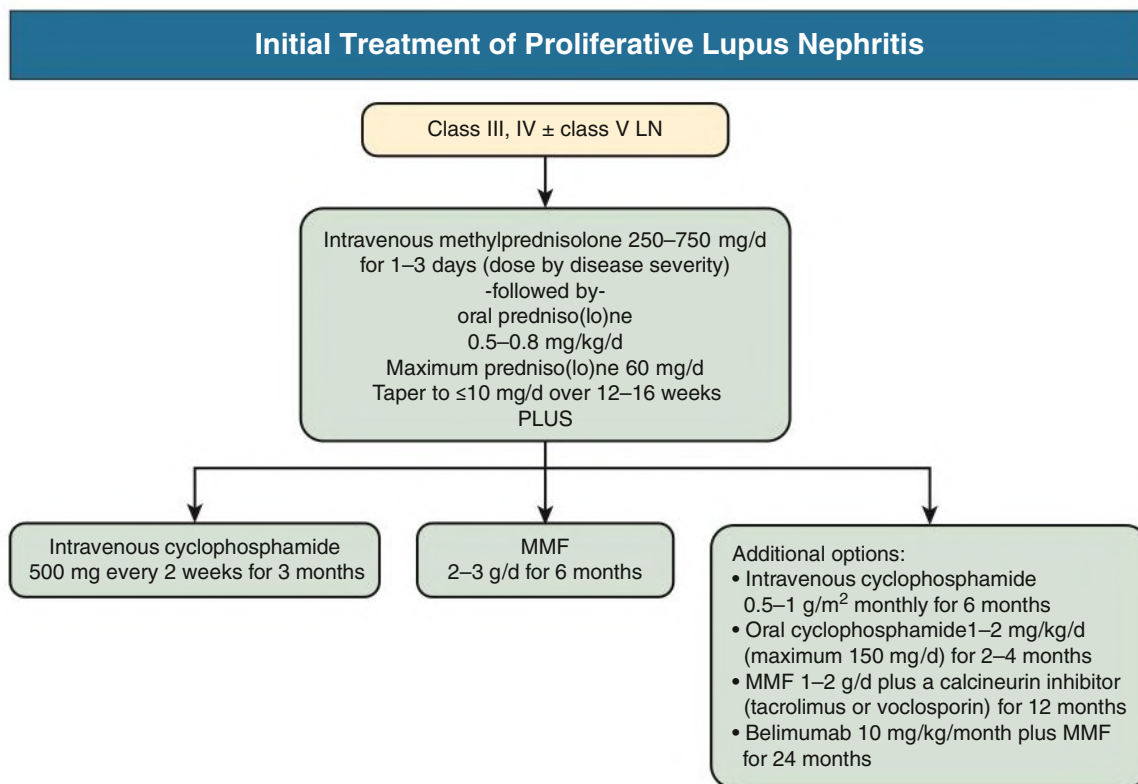


Fig. 27.9 Approach to the initial treatment of proliferative lupus nephritis (LN) AZA, Azathioprine; MMF, mycophenolate mofetil.

have been suggested but not confirmed.⁶² However, a retrospective analysis of a Korean cohort showed similar remission rates for MMF and cyclophosphamide but more relapses and a higher incidence of ESKD in the MMF group.⁶³

Guidelines for immunosuppressive therapy from the nephrology and rheumatology societies have been updated.^{53,54} Our approach to the initial treatment of proliferative LN is given in Fig. 27.9.

Other Immunosuppressive Strategies

Azathioprine (AZA) has been used in combination with glucocorticoids for the induction of remission in proliferative LN. A randomized controlled trial (RCT) comparing AZA with cyclophosphamide found no difference in nonsustained doubling of initial serum creatinine, but more relapses, worse kidney function, and more chronicity on repeat biopsy in the AZA group.⁶⁴ Although AZA is not recommended as first-line therapy, it remains an option when MMF or cyclophosphamide are unavailable, undesirable, or contraindicated (including during pregnancy).

CNIs in combination with glucocorticoids or as part of a three-drug regimen consisting of prednisone, MMF, and a CNI have been used primarily in Asia for the initial treatment of LN. In the largest Chinese three-drug combination study, tacrolimus plus MMF and glucocorticoids were compared with high-dose cyclophosphamide plus glucocorticoids. The CNI-based regimen resulted in a greater complete kidney remission rate than cyclophosphamide at 6 months for all LN classes.⁶⁵ A CNI-based approach was also tested in a phase II trial in a highly diverse patient population that compared MMF, prednisone, and placebo with MMF, prednisone, and voclosporin.⁵² The addition of voclosporin resulted in significantly more complete kidney responses at 6 and 12 months. In a phase III study (AURORA) of 357 patients with class III, IV, or V LN, the additive benefit of voclosporin was confirmed.⁶⁶ Long-term preservation of kidney function and

improvement in kidney histology in CNI-treated LN patients will be addressed to some extent through the extension phase of AURORA.

The AURORA trial design did not distinguish between induction and maintenance. Other than a tapering of low-dose glucocorticoids, patients remained on the same medications and doses for 12 months.

Biologic Agents

Rituximab, an anti-CD20 monoclonal antibody that depletes B cells, was tested in the LUNAR trial, a randomized prospective study of patients with active LN. LUNAR failed to detect a difference in complete kidney responses at 1 year when rituximab was added to MMF and glucocorticoids compared with MMF and glucocorticoids alone.⁶⁷ In subsequent clinical trials, other biologics, such as Abatacept, an inhibitor of T- and B-cell costimulation, an anti-interleukin 6 monoclonal antibody, and an anti-TWEAK monoclonal antibody, also failed to improve the proportion of LN patients achieving a complete kidney response.^{57,68,69}

Despite the trial's failure, follow-up of LUNAR participants beyond 12 months suggested that the rituximab and placebo groups need more time to show differences.⁶⁷ Additionally, a post hoc analysis of LUNAR showed that complete kidney responses correlated with complete peripheral B cell depletion that occurred early and was sustained.⁷⁰ Using this information to inform trial design, the humanized anti-CD20 monoclonal antibody obinutuzumab was tested in NOBILITY, a 2-year, phase II LN RCT. Like rituximab, obinutuzumab depletes peripheral B cells, but compared with rituximab, depletion is more complete and sustained, there is more antibody-dependent cytotoxicity, and there is less reliance on complement-dependent cytotoxicity.^{71,72} NOBILITY compared obinutuzumab to placebo on a background of MMF and prednisone. Besides being longer than LUNAR, overall glucocorticoid dosing was less, and there was little separation into induction and maintenance phases in NOBILITY,

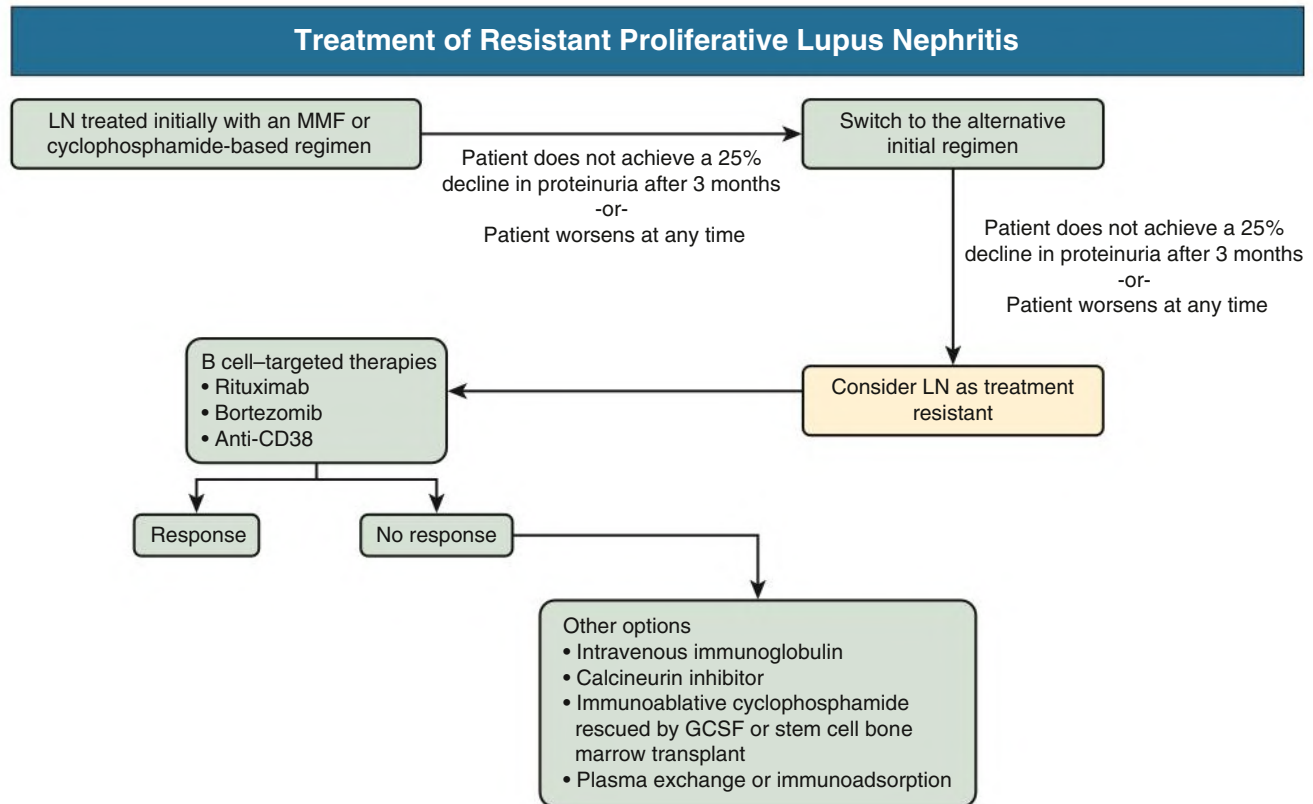


Fig. 27.10 Treatment of resistant proliferative lupus nephritis (LN). GCSF, Granulocyte colony-stimulating factor; MMF, mycophenolate mofetil.

other than the last infusion of obinutuzumab or placebo at month 6. Significantly more patients who received obinutuzumab achieved complete kidney response at week 76 and week 104 compared with placebo.^{73,74} Adverse events were comparable between the two groups.

Belimumab, a monoclonal antibody that targets the B-cell growth and survival factor BAFF (also called BLyS), was tested in a large ($n = 448$) phase III trial (BLISS-LN).⁴⁹ BLISS-LN was a 2-year study that allowed investigators to choose background immunosuppression, and the primary endpoint was primary efficacy renal response (PERR), whereas complete kidney response was a secondary endpoint. For PERR, the urine protein to creatinine ratio was 0.7 or less compared with less than 0.5 for complete response, and estimated GFR (eGFR) had to be at least 60 mL/min/1.73 m² and no more than 20% below its preflare value, whereas complete response required an eGFR of at least 90 mL/min/1.73 m² and no more than 10% below its preflare value. Belimumab was added to MMF plus glucocorticoids followed by MMF or to low-dose cyclophosphamide plus glucocorticoids followed by azathioprine. Glucocorticoids had to be tapered to 10 mg/day or less by week 24 and could only be increased briefly to treat a non-renal SLE manifestation, and not at all between week 76 and week 104 or the treatment was considered a failure. At 104 weeks, patients who received belimumab had significantly more PERRs and complete kidney responses than those given placebo.⁴⁹ Belimumab-treated patients had significantly fewer kidney-related events and better preservation of kidney function over the 2 years of the trial. Adding belimumab to background immunosuppression did not increase adverse events compared with placebo.

In summary, although the early experience with biologics for the treatment of LN was discouraging, longer trial duration, a blurring of the line between induction and maintenance therapy, and lower exposure to glucocorticoids allowed for successful trials. B cell-targeted

biologics do not appear to act quickly, consistent with the fact that B cells do not produce autoantibodies but are precursors to autoreactive plasma cells and plasmablasts. Antibodies to type 1 interferon and to complement system components are currently being assessed for the treatment of LN.

Renal Response to Initial Therapy

LN patients who achieve a complete kidney response during treatment generally have 5-year patient and kidney survival of more than a 90% compared with only 69% (patient) and 45% (kidney) for those not achieving complete remission.¹¹ Partial responses are also associated with improved outcomes compared with no response. It is also clear that complete kidney responses take time. About half of LN patients treated with MMF or cyclophosphamide achieve a complete or partial response by 1 year, with an additional 25% by 2 years.⁷⁵ To avoid prolonged exposure to inflammatory injury, a 25% reduction in proteinuria should be targeted during the first 3 months of therapy and proteinuria should be reduced by 50% by 6 months.⁵³ At 12 months, patients would ideally have a proteinuria level less than 0.7 g/day. Using these metrics, consider switching medications if a patient has not demonstrated a 25% decline in proteinuria at 3 months. If the patient had been started on an MMF-based regimen, a switch to a cyclophosphamide-based regimen is appropriate, or vice versa. After another 3 months, if the patient has not shown a 25% decline in proteinuria, they could be considered as treatment resistant, at least to usual therapies (Fig. 27.10). A critical caveat before calling a patient treatment resistant is to verify adherence to therapy.⁷⁶

Resistant Proliferative Lupus Nephritis

There is no high-quality RCT evidence to suggest an optimal approach to patients with resistant LN. In uncontrolled trials, cohort studies, and

observational studies in resistant LN, rituximab was added to, or given after, intense conventional immunosuppression and converted 40% to 50% of nonresponders to complete responders.^{77,78} Rituximab also has been combined with belimumab for refractory LN.⁷⁹ Plasma cells have been targeted in a small number of resistant LN patients with bortezomib or anti-CD38 antibodies with some success.^{80,81}

Other possible beneficial approaches tested in small series of patients with resistant LN include plasma exchange, IV immunoglobulin, and CNIs.^{82,83,84} Infusion of allogeneic mesenchymal stem cells derived from umbilical cord or bone marrow has been tested in refractory LN.⁸⁵ At 12 months, about 23% of patients had a complete renal response, but 40% of patients were considered to have treatment failures. Finally, for patients with life-threatening resistant disease, small pilot studies have tested total lymphoid irradiation and immunoablation by high-dose cyclophosphamide and antithymocyte globulin, with or without autologous stem cell reconstitution. These approaches have led to sustained treatment-free remissions, but toxicity and treatment-related mortality are considerable.⁸⁶

Proliferative Lupus Nephritis: Subsequent Therapy

The high adverse event rate from intense cytotoxic treatment can be mitigated with no loss of efficacy if cyclophosphamide is discontinued after 3 to 6 months and drugs such as azathioprine or MMF are substituted in for long-term disease control.⁸⁷ The concept of induction followed by maintenance therapy has been challenged (see earlier), and LN is thought of as a chronic disease for which preservation of kidney function is the most important outcome.

MMF is the drug of choice for long-term immunosuppression after initial therapy with cyclophosphamide or MMF, based on a trial that demonstrated MMF was superior to azathioprine at preventing renal flares, preserving kidney function over 3 years, and delaying progression to ESKD.⁶¹ If MMF is not tolerated, or in specific circumstances such as pregnancy, azathioprine or a CNI may be used for long-term immunosuppression.

The optimal duration of immunosuppression in patients with LN is unknown. Patients generally remain on immunosuppression for several years, and the longer the duration of therapy, the less chance of flare.⁸⁸ The concern with withdrawing therapy too early is that LN will flare (relapse) and that every flare will increase the risk of progressive kidney disease. On the other hand, continuing immunosuppression increases the risk for adverse events.

Protocol kidney biopsies may help decide whether immunosuppressive therapy can be safely withdrawn. In patients with proliferative LN, a protocol biopsy done after 36 months of immunosuppression and at least 12 months of complete clinical renal remission showed that 30% of patients still had ongoing immunologic activity. Immunosuppression was withdrawn from all patients in clinical remission and LN relapsed almost exclusively in patients with a residual histologic activity index of at least 2.⁸⁹ Using a similar approach,⁹⁰ if a 42-month protocol biopsy showed a histologic activity index of 0, immunosuppression was discontinued, but if the activity index was 1 or greater, immunosuppression was continued. After 24 months, biopsies were performed again in the patients who remained on immunosuppression, and if the activity index was 0, the immunosuppressives were discontinued. Using this approach, the observed flare rate was fivefold less than that expected after discontinuation of immunosuppression based only on clinical data. Thus, performing a kidney biopsy can help inform the decision to withdraw immunosuppression. It is conceivable that lower cumulative immunosuppression may be sufficient for some patients, perhaps those with no histologic activity on kidney biopsy after 1 or 2 years of treatment.

The long-term management of immunosuppression for patients with persistent proteinuria who have achieved only partial remission is even less certain. Persistent proteinuria may be because of ongoing inflammatory disease activity or chronic damage and nephron loss. It has been found that a proportion of patients with persistent proteinuria no longer has histologic activity on kidney biopsy.⁹¹ Such patients may not need ongoing immunosuppression.

Membranous Lupus Nephropathy

Membranous nephropathy is often diagnosed together with proliferative forms of LN. In these patients, treatment is directed at the proliferative component. Alternatively, the combination of low doses of glucocorticoids, MMF, and a CNI has shown considerable success in an Asian cohort of mixed membranous and proliferative LN.⁹²

Immunosuppressive therapy should be used for patients with nephrotic syndrome and/or impaired kidney function. The only RCT in patients with pure class V LN was small and compared cyclophosphamide or cyclosporine with glucocorticoids alone.⁹³ The patients studied had preserved kidney function and a mean proteinuria of almost 6 g/day. Complete and partial remissions were more frequent in the cyclophosphamide and cyclosporine-treated patients. Cyclosporine induced remission more rapidly than the other drugs, but LN relapses were fewer in the cyclophosphamide group. Two trials of MMF versus cyclophosphamide as initial therapy of LN included 84 patients with pure membranous nephropathy, among others.⁹⁴ Remissions, relapses, and courses were similar in the class V patients treated with MMF and cyclophosphamide. Pure class V LN with subnephrotic proteinuria can be treated with renoprotective and antiproteinuric therapies and immunosuppression dictated by extrarenal lupus symptoms. Our approach to treatment of class V LN is outlined in Fig. 27.11.

Long-Term Monitoring of Lupus Nephritis Patients

The relapse (flare) rate for LN ranges from 35% to 60%, depending on the population studied, the criteria for relapse, and the maintenance therapy used.⁹⁰ In LN patients in remission, quarterly monitoring (including blood pressure, kidney function, proteinuria, urinary sediment, and serum C3 and C4) is recommended, with anti-dsDNA measured at least biannually. Changes in serology alone do not warrant therapeutic action, although some investigators report lower flare rates when modest preemptive treatment is given to such patients.⁹⁵ At the very least, patients should be watched even more closely for flare. If LN does flare, a repeat biopsy should be considered if a change in histologic class is suspected. Patients with proliferative LN tend to remain proliferative at flare, but patients with class II and V not infrequently develop a proliferative component.

ANTIPHOSPHOLIPID ANTIBODY SYNDROME, ATHEROSCLEROSIS, AND PREGNANCY IN LUPUS NEPHRITIS

Intrarenal thrombosis caused by the antiphospholipid antibody syndrome is found in 30% of patients with SLE and may occur in the presence or absence of LN (see Chapter 29). The mainstay of treatment for antiphospholipid nephropathy is anticoagulation plus an antimalarial, although immunosuppressive agents have also been used. Vitamin K antagonists such as warfarin (VKAs) are the anticoagulants of choice for arterial thromboses; the choice between VKAs and direct oral anticoagulants for patients with venous thrombosis remains controversial.⁹⁶

Patients with lupus have markedly elevated risk for atherosclerotic cardiovascular disease and a greater atherosclerotic plaque burden compared with age-matched controls.^{97,98} Reduction of atherosclerotic

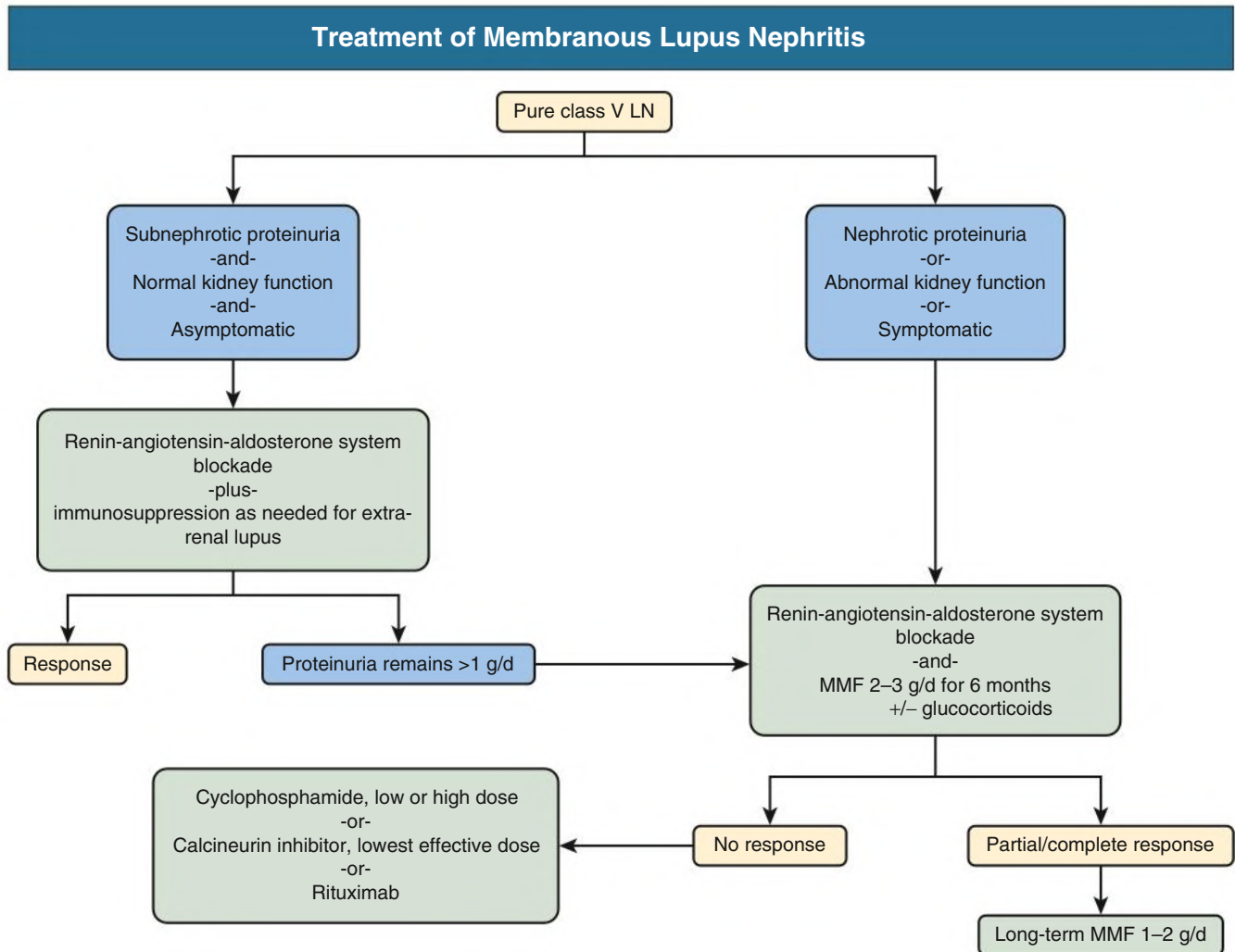


Fig. 27.11 Treatment of membranous lupus nephritis (LN). MMF, Mycophenolate mofetil.

risk should focus on blood pressure control (goal of <130/80 mm Hg), use of statins and hydroxychloroquine,⁹⁹ and reduction of inflammatory disease activity.

The effects of SLE and LN on pregnancy and fetal outcomes and the effects of pregnancy on LN activity are discussed in [Chapter 45](#).

END-STAGE KIDNEY DISEASE AND KIDNEY REPLACEMENT THERAPY

Although survival of LN patients on dialysis was noted to be similar to patients with other causes of ESKD,¹⁰⁰ a study from Australia suggested

worse survival.¹⁰¹ LN patients who received a kidney transplant had significantly better survival than those who did not,¹⁰² suggesting that kidney transplantation is the kidney replacement modality of choice for LN patients with ESKD. Extrarenal lupus is often quiescent by the time patients reach ESKD, but those with active extrarenal disease may still require immunosuppression. It has been suggested that patients with LN defer transplantation for 3 to 6 months to allow SLE to become inactive; however, an increased risk for allograft failure was found when LN patients waited more than 3 months before transplantation.¹⁰³ Recurrent LN after kidney transplantation may occur but is generally not the cause of allograft loss.¹⁰⁴

SELF-ASSESSMENT QUESTIONS

1. A 21-year-old African-American woman presents for initial therapy of LN. She was diagnosed with SLE 1 year ago, when she was found to have alopecia, malar rash, and arthralgias with ANAs and dsDNA antibodies. At that time, urine sediment was unremarkable, serum creatinine was 0.6 mg/dL, and she had no proteinuria. She was treated with low-dose glucocorticoids and hydroxychloroquine, with resolution of her symptoms. Over the past month, she noticed foamy urine and mild ankle swelling. Evaluation showed blood on her urine dipstick, 24-hour urine with 4 g protein, serum creatinine of 1 mg/dL, and a rising dsDNA titer with both C3 and

C4 now newly decreased. A kidney biopsy revealed class IV LN, without crescents or glomerular capillary necrosis. The patient wants children in the future. Considering options for initial therapy for this patient's LN, which of the following statements is *incorrect*?

- She could be treated with MMF and glucocorticoids.
- She could be treated with a calcineurin inhibitor in combination with MMF and glucocorticoids.
- She could be treated with low-dose cyclophosphamide and receive ovarian protection.
- She could be treated with belimumab alone.

2. A 24-year-old Hispanic woman has been treated for the last 6 months with MMF 3 g/day and a tapering dose of glucocorticoids for active focal proliferative class III LN. Her initial proteinuria decreased from 3.4 g/day to 1.2 g/day, urine sediment became inactive, serum creatinine decreased from 1.6 to 0.9 mg/dL over the 6 months of therapy, anti-dsDNA titer declined, and serum complement values returned to normal. She has tolerated her treatment well and reports no side effects. What should be done with her immunosuppression?
- A. Continue the current dose of MMF, follow patient for a complete renal remission, and when achieved, reduce MMF to 2 g/day and continue for at least 36 months of total immunosuppressive therapy, with ongoing monitoring for any evidence of LN relapse.
 - B. Change to intravenous cyclophosphamide at 500 mg every 2 weeks for 6 doses and then administer azathioprine at 2 mg/kg daily.
 - C. Add a calcineurin inhibitor to her current regimen.
 - D. Add rituximab to her current regimen.
3. A 32-year-old White woman with a past history of LN wants to become pregnant and is concerned about flares during the pregnancy. Which of the medications listed should NOT be used during pregnancy?
- A. Glucocorticoids
 - B. Calcineurin inhibitor
 - C. Hydroxychloroquine
 - D. Azathioprine
 - E. Mycophenolate mofetil
4. A 28-year-old White man had been diagnosed 4 years ago with class IV LN by kidney biopsy. He was treated with MMF and glucocorticoids, and his kidney function improved to an eGFR of 89 mL/min/1.73 m². Proteinuria fell from 2.7 g/day to 950 mg/day. His anti-dsDNA antibody titer fell by 75% and his C3 complement increased to the normal range, but C4 remained low. He has been on MMF 2 g/day for 3 years. He has been on 5 mg of prednisone every other day for 2 years. He has had no extrarenal signs or symptoms of lupus for over 3 years. He wants to know if he needs to stay on immunosuppression or can safely come off. What should be done?
- A. Continue his MMF and prednisone until the anti-dsDNA antibody is no longer detectable and then stop immunosuppression.
 - B. Order a kidney biopsy to determine whether there is any residual histologic activity.
 - C. Continue his MMF and prednisone until his C4 level is in the normal range and then stop immunosuppression.
 - D. Continue his MMF and prednisone until his proteinuria is less than 250 mg/day.

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